Project Introduction: A Thymus-Anchored Map of Tertiary Lymphoid Structures Across Multiple Autoimmune Diseases

Topic: Computational Biology & Bioinformatics

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Abstract

1.1 Introduction

Autoimmune diseases can be defined by the hyperactivation against one's own tissues. This often results in the formation of ectopic Tertiary Lymphoid Structures (TLS). Standard-of-care treatments currently rely heavily on broad systemic immunosuppressants (corticosteroids, Rituximab) and surgery. Although these drugs control symptoms, they often harm healthy peripheral cells.

While recent single-cell RNA sequencing (scRNA-seq) atlases have compared peripheral diseases such as Rheumatoid Arthritis (RA) and Systemic Lupus Erythematosus (SLE), they have historically omitted the central thymic pathology of Myasthenia Gravis (MG) from cross-disease integrations. Because 80% of Myasthenia Gravis begins in the thymus, i.e. that is primary for central tolerance, it provides a starting point to study how these rogue immune structures are built.

1.2 Aim

1. To determine if peripheral autoimmune diseases (SLE, RA, Sjögren's Syndrome, MG and Graves Disease) utilise the same central immune structure as Myasthenia Gravis to create Tertiary Lymphoid Structures.
2. Use this shared architecture as a subtractive baseline to isolate therapeutic targets using Query approaches for each organ.

1.3 Methods

1. Data Assembly: Collect scRNA-seq datasets for MG (Thymus), RA, SLE, Sjögren's Syndrome, Graves' Disease, and healthy tonsil tissue (which will be the statistical control group).
2. TLS Isolation: Identify the cell interactions driving MG so that we can extract the shared immune structure of the 5 autoimmune diseases.
3. Cross-Disease Projection: Apply the MG immune structure to the other disease datasets using AUCell scoring to calculate the proportion of the shared immune response.
4. Stromal Extraction: Computationally subtract the shared immune response to isolate non-immune stromal targets (e.g., thymic epithelial cells, synovial fibroblasts) that uniquely cause local tissue damage.
5. Pharmacogenomic Mapping: Query the isolated stromal targets against chemogenomic databases (DGIdb, ChEMBL) to identify medication/drugs that can be localised to these targets.

References:

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